

Bank Customer Data analysis using SQL

Analyzing Customer and Transaction
Data Using SQL





Project Overview

- Analyzes banking customer and transaction data using SQL to solve business problems and generate insights.



Objectives

- Practice SQL
- Analyze customer & transaction data
- Build business reports
- Strengthen data analysis skills



Problem Statement

The objective of this project is to analyze customer and transaction data from a banking database using SQL to answer business questions and generate actionable insights.

Print customer Id, customer name and average account_balance maintained by each customer for all of his/her accounts in the bank.

QERY:

```
select b.customer_id, b.customer_name,avg(d.Balance_amount) as average_account_amonut
from BANK_CUSTOMER as b
join Bank_Account_Details as d
on b.customer_id=d.customer_id
group by b.customer_id, b.customer_name
order by average_account_amonut ;
```

RESUL:

Results Messages			
	customer_id	customer_name	average_account_amonut
1	123008	Robin	0
2	123006	Noah	151666
3	123005	Jacob	300000
4	123003	Harry	750000
5	123007	Charlie	1845000
6	123002	George	3950000
7	123004	Jack	4077500
8	123001	Oliver	4800000



Print customer_id , account_number and balance_amount , condition that if balance_amount is nil then assign transaction_amount for account_type = "Credit Card"

QERY:

```
select d.customer_id,d.account_number,t.Transaction_amount as balance_amount
from Bank_Account_Details d
join BANK_ACCOUNT_TRANSACTION t
on d.account_number=t.account_number
where d.Balance_amount=0 and d.account_type = 'credit Card'
```

RESUL:

Results		Messages	
	customer_id	account_number	balance_amount
1	123007	9000-1700-7777-4321	-5000.00
2	123007	9000-1700-7777-4321	-8000.00
3	123007	9000-1700-7777-4321	-2500.00
4	123006	5800-1700-9800-7755	-9000.00



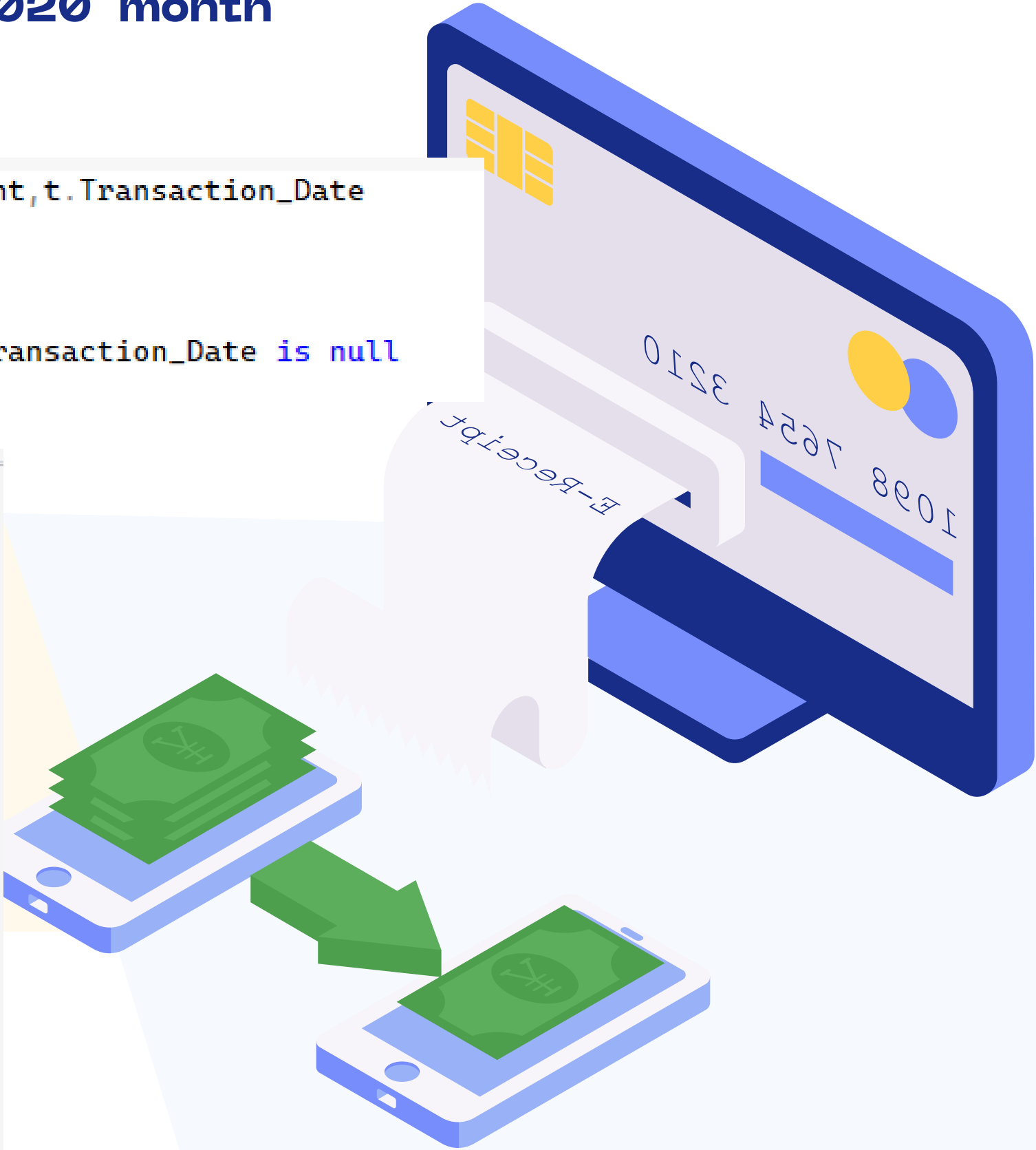
Print all of the customer id, account number, balance_amount, transaction_amount , Transaction_Date from bank_customer,Bank_Account_Details and bank_account_transaction tables where excluding all of their transactions in march, 2020 month

QERY:

```
|
select a.customer_id,d.account_number,d.balance_amount,t.transaction_amount,t.Transaction_Date
from bank_customer a
left join Bank_Account_Details as d on a.customer_id=d.customer_id
left join BANK_ACCOUNT_TRANSACTION t on d.Account_Number=t.Account_Number
where t.Transaction_Date not between '2020-03-01' and '2020-03-31' or t.Transaction_Date is null
```

RESUL:

	customer_id	account_number	balance_amount	transaction_amount	Transaction_Date
1	123001	4000-1956-3456	200000	-2000.00	2020-01-13
2	123001	5000-1700-3456	9400000	NULL	NULL
3	123002	4000-1956-2001	400000	-4000.00	2020-02-14
4	123002	4000-1956-2001	400000	-1600.00	2020-01-19
5	123002	4000-1956-2001	400000	-3000.00	2020-04-24
6	123002	5000-1700-5001	7500000	NULL	NULL
7	123003	4000-1956-2900	750000	NULL	NULL
8	123004	5000-1700-6091	7500000	40000.00	2020-02-19
9	123004	4000-1956-3401	655000	8000.00	2020-01-19
10	123006	5800-1700-9800-7755	0	-9000.00	2020-04-13
11	123007	5000-1700-9800	355000	NULL	NULL
12	123007	4000-1956-9977	7025000	50000.00	2020-01-16
13	123007	4000-1956-9977	7025000	10000.00	2020-02-16
14	123007	4000-1956-9977	7025000	60000.00	2020-04-18
15	123007	4000-1956-9977	7025000	49000.00	2020-06-18
16	123007	9000-1700-7777-4321	0	-5000.00	2020-02-17
17	123007	9000-1700-7777-4321	0	-2500.00	2020-04-21
18	123007	5900-1900-9877-5543	0	NULL	NULL
19	123008	5000-1700-7755	0	NULL	NULL



Display all saving account holders have "Add-on Credit Cards" and "Credit cards"

QERY:

```
SELECT
    s.Customer_id
FROM Bank_Account_Details s
JOIN Bank_Account_Relationship_Details c
ON s.Account_Number=c.Linking_Account_Number
JOIN Bank_Account_Relationship_Details ac
ON c.Account_Number=ac.Linking_Account_Number
WHERE
s.Account_type='SAVINGS'
AND c.Account_type='Credit Card'
AND ac.Account_type='Add-on Credit Card';
```

RESUL:

Results		Messages		
	Customer_id	Account_Number	Account_type	balance_amount
1	123001	4000-1956-3456	SAVINGS	198000.00
2	123002	4000-1956-2001	SAVINGS	396000.00
3	123002	4000-1956-2001	SAVINGS	398400.00
4	123002	4000-1956-2001	SAVINGS	394000.00
5	123002	4000-1956-2001	SAVINGS	397000.00
6	123002	4000-1956-2001	SAVINGS	423000.00
7	123004	4000-1956-3401	SAVINGS	663000.00
8	123005	4000-1956-5102	SAVINGS	293500.00
9	123006	4000-1956-5698	SAVINGS	446000.00
10	123007	9000-1700-777...	Credit Card	-5000.00
11	123007	9000-1700-777...	Credit Card	-8000.00
12	123007	9000-1700-777...	Credit Card	-2500.00
13	123006	5800-1700-980...	Credit Card	-9000.00



Thank You

GitHub Repository:

https://github.com/dhanushj1139/Banking_Customer_Data_analysis_using_SQL

